

(a) Shared representations

Human video

Same task, different timing



Cached tokens

512-D / token



128-D

frame embeddings

$f\theta$

Robot video

Same task, different timing



Cached tokens

512-D / token



128-D

frame embeddings

$f\theta$

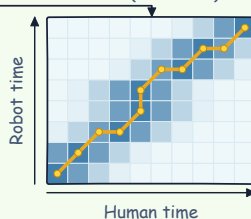
Shared weights

Video context · cached token sets · shared adapter

(b) Monotonic matching

Cosine similarity

matrix (schematic)



Human time

Two-way DTW

R to H H to R
Complete sequences · offline

(c) Filtered correspondences

Filtering criteria



Similarity

Cycle consistency

Match margin

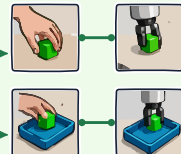
ALL must pass

Cosine ≥ 0.50
Cycle ≤ 2 frames*
Margin ≥ 0.01

Retained frame pairs

Human

Robot

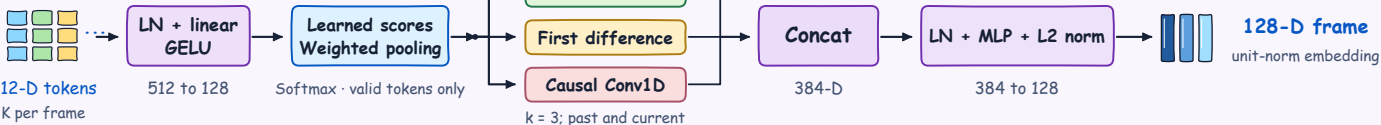


Rejected candidate



(d) Shared adapter: learned pooling + causal context

Frozen at inference



* Cycle error in sampled robot frames. Margin: competitors >3 human frames away; none: reject.